Shader cheat sheet

WEEK 9 / UNREAL 5.7

Finish your existing shader. Change one control; compare under the same lighting.

CHOOSE THE EFFECT

Height blend — reshapes a transition between surfaces; does not move vertices.

POM — shifts texture sampling for apparent depth; silhouette and collision stay unchanged.

WPO — moves existing vertices. Check mesh density, deformation masks and shading.

Flow — RG supplies direction; a scalar mask controls where/how much motion contributes.

CHOOSE THE CONTROL SCOPE

Material instance — affects every object using it. Duplicate **and assign** for a separate variant.

Dynamic instance (MID) — assign it, then update that same instance at runtime.

Collection (MPC) — shared by every material reading the collection.

Primitive data (CPD) — belongs to one component. **Static switches** select compiled variants.

MASKS, MOTION & LAYERS

Lerp(A, B, mask): 0 selects A; 1 selects B. **Saturate** clamps to 0–1.

UV × tiling + offset: more tiling makes repeated features smaller. UV motion does not move the mesh.

Direction mask: compare normals and a normalized direction in the same coordinate space.

Native Material Layers: layers define surface attributes; blends control how layers combine.

NO VISIBLE CHANGE?

1. Is the material assigned to the correct mesh or decal?

2. Is the parameter override enabled and its branch active?

3. Are channel, color space and parameter scope correct?

4. For decals: inspect projection bounds, Receives Decals and the receiver's Decal Response.

5. Match camera, light and exposure. Record original values before experimenting.

OPTIONAL GALLERY CHECKS

Ref 08 · Height blend: MI_08_HeightBlend → Blend Contrast: 1 vs 4; keep paint fixed.

Ref 11 · Shared controls: SharedRoughness: .2 vs .8. Reset; then test one component's CPD RGB 0–2.

Ref 10 · Decals: MI_10_RoughnessDecal → Roughness: .03 vs .8; judge reflected light.

Ref 07 · POM: Near Height: 0 vs .04 at a close view. Keep steps fixed; inspect the silhouette.

Reference IDs match the slides; wall numbers show the gallery route. Restore temporary tests.

SHADER HOMEWORK
Submit a video showcasing your shader/material, **with a breakdown explaining how it works**. Connect the graph's inputs, masks and logic to the visible result.

More detail + optional experiment record online